CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2013

EC404 Embedded Systems

Date: 19.11.2013, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) What is design metrics? Define four different design metrics of embedded systems. **[03]**

(b) Define following terms: **[04]**

i) Task Control Block, ii) Inter Process Communication

Q - 2 (a) Explain Shared data Problem and its solution using Semaphore. **[07]**

(b) Discuss the following content of I2C bus: **[07]**

i) structure, ii) electrical interface, iii) format of an I2C data frame, iv) application interface.

OR

Q - 2 (a) Explain Processor technology which used for designing Embedded Systems. Also compare all three major category named i) General Purpose, ii) Single Purpose, iii) Application Specific. **[07]**

(b) Interface EEPROM 2708 with 8051 microcontroller. Also write a 8051 program to write and read the data from 2708 using I2C protocol. **[07]**

Q - 3 (a) Write short note on CAN bus. **[07]**

(b) What is RTOS? How does it differ from conventional OS? Describe three function and their activities provided by RTOS services. **[07]**

OR

Q - 3 (a) With neat diagram, Interface LPC2148 with 4*4 Keypad, also write a C program to detect the pressed key. **[07]**

(b) With the help of example, Explain following RTOS scheduling **[07]**

i) Cooperative Scheduling of Ready Tasks using Precedence Constrains,

ii) Preemptive scheduling.

SECTION – II

Q - 4 (a) What are the main phases of a design review? **[03]**

(b) Briefly describe the differences between the waterfall and spiral development models. **[04]**

Q - 5 (a) Interface LCD with LPC2148 using only 6 I/O pins. Write a program to display "CHARUSAT IS BEST". **[06]**

(b) Compare RICS and CISC processor. Why the RISC is more suitable for embedded system like network processor? **[04]**

(c) With help of suitable example explain all possible case in stack addressing. **[04]**

OR

Q - 5 (a) What is advantage and disadvantage of pipeline? With neat sketch explain 3-stage pipeline and its drawback. **[06]**

(b) With help of suitable example explain data processing instruction of ARM processor. **[04]**

(c) Discuss the importance of "time-to-market" design metric with the help of example. **[04]**

Q - 6 (a) Write a Technical note on Quality Assurance in Embedded System Design. **[07]**

(b) Differentiate general purpose processors and DSP processors. **[04]**

(c) Differentiate ARM mode and thumb mode. **[03]**

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